Variational assimilation of ocean surface using a truncature-acknowledging reduced-gravity model

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Keypoints

- Enhancing reduced-gravity model reconstruction in the context of data assimilation by acknowledging the truncation error
- Introducing a physics-derived regularization term for the cost function to enforce multilayer QG dynamics
- Improving the robustness of the model by introducing a deformation radius correcting parameter in the control vector

1 Introduction

In physical oceanography, the majority of available data is acquired by satellites that cannot measure the interior ocean state. The recent launch of the Surface Water and Ocean Topography (SWOT) mission (Fu et al., 2024) has paved the way for sea surface reconstruction with unprecedented resolution. However, this satellite covers the entire surface of the globe in 21 days. Although most parts of the globe are visited multiple times within this 21d-period, many gaps remain in the collected data. The reconstruction of sea-surface dynamics through partial observations has been investigated for a long time. In the late 1980s, physics-motivated flow reconstruction methods were developed, either focusing on surface (Cong & Ikeda, 1995; Cong et al., 1998; Cooper & Haines, 1996; Holland & Malanotte-Rizzoli, 1989; Mey & Robinson, 1987) or subsurface (Haines, 1991; Hurlburt, 1986) flow. In the 1990s, statistical methods such as optimal interpolation (Rienecker & Miller, 1991; Traon et al., 1998) emerged, forming the basis of the DUACS product which remains a widely used real-time gridded ocean product. This interpolation method uses only observation data to reconstruct surface ocean state.

To make use of theoretical models of ocean surface dynamics, data assimilation (DA) techniques started to be applied to the ocean system. In order to reduce reconstruction errors, DA combines both simulated and observed data while accounting for their specific sources of error (Lahoz et al., 2010). A wide variety of methods have been developed and applied to the reconstruction of surface ocean dynamics (Martin et al., 2025). Simulated system state is usually propagated using so-called reduced-order models. These models rely on simplified frameworks to emulate ocean system dynamics, allowing to perform efficiently complex DA algorithms. Around the midlatitudes, a simplified framework of choice is

the quasi-geostrophic (QG) theory. This framework, developed in the 1950s (Charney & Phillips, 1953; Charney et al., 1950), describes the behavior of flows around the geostrophic balance through the material conservation of potential vorticity (PV) by the geostrophic flow. Such theory has paved the way for many simplified numerical models, whether atmospheric or oceanic (Held et al., 1995; Phillips, 1951). Although it describes 3D dynamics, it can be discretized by stacking layers of fluid with constant vertical properties, yielding the multilayer QG framework (Vallis, 2006). The ability of such framework in reconstructing sublayer flow has been widely investigated in the late 1980s (Haines, 1991; Hurlburt, 1986), but satisfying reconstructions are achieved after numerous surface observations have been assimilated into the models. Regarding surface flow reconstruction, assimilation into a two-layer QG model has been experimented by Cong et al., 1998 and showed encouraging results. However, both the lack of subsurface direct observations (and thus of precise estimation for initial sublayer flow state) and the motivation of reconstructing only surface flow have motivated the use of reduced-gravity (RG) QG model in data assimilation. This framework is directly derived from the multilayer one by considering deep ocean flow to be at rest. This additional simplification offers an evolution equation for the surface flow without requiring any information on the subsurface state. These specific characteristics naturally led to the choice of the RG framework for modeling surface dynamics.

Using a RG model, Ubelmann et al., 2015 presented the so-called dynamical interpolation method that consist in refining the optimal interpolation reconstruction by using such model to benefit from its ability to render non-linear dynamics. This method has been able to improve the reconstruction of the flow (Ubelmann et al., 2016). In order to account for any source of error in the RG model, the VarDyn framework (Le Guillou et al., 2023, 2025) introduces an external forcing into the RG advection equation. Such forcing is estimated through variational assimilation to approach the rich and complex dynamics of the ocean surface. However, in both methods remains the approximation that the flow is at rest below the ocean surface.

Motivated by a rewriting of the top layer QG equations, we tackle truncation error through a parameterization of the sublayer stream function. We introduce an external forcing in the RG equations which represents the advection of sublayer stream function by the surface flow. In a context of partial surface observations, variational assimilation is then performed to find the sublayer field that allows the most accurate reconstruction of the surface flow. As our method provides a direct estimation for sublayer stream function, we are also able to replace the usual background term of the loss function by a physics-based regularization enforcing multilayer QG dynamics of the surface flow. Method validation is then performed through two experiments: comparison to a three-layer QG model to monitor truncation-only error mitigation, and comparison to a primitive equation simulation to assess the model's reconstruction capabilities of a flower with richer dynamics.

2 Method

2.1 The multilayer quasi-geostrophic equations

QG dynamics relies on the material conservation of PV by geostrophic flow. Both the PV and the geostrophic flow can be expressed using the stream function which, through the inversion of the elliptic operator mapping stream function to PV, allows recovering the

geostrophic flow from the PV itself (Hoskins et al., 1985). Multilayer QG, the vertical discretization of the continuously stratified QG equation, is obtained by stacking layers of fluid with constant vertical properties (Pedlosky, 1987; Vallis, 2006) inside which PV is conserved. In the following we will recall these equations in a N-layer fluid. Using $\underline{\cdot}$ to represent layered fields ($\underline{\chi} = [\chi_1, \dots, \chi_N]^\top$), we denote

- $\underline{\rho}$ the reference layer densities,
- $\underline{H}$ the reference layer thickness,
- $\underline{g}$ the reduced gravities ($g_i = g\left(\rho_i - \rho_{i-1}\right)/\rho_1$, with the convention $g_1 = g$),
- $\underline{\psi}$ the stream function vector
- $\underline{q}$ the PV vector

Introducing the horizontal gradient $\boldsymbol{\nabla} = \left[\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y}\right]^\top$ along with the perpendicular horizontal gradient $\boldsymbol{\nabla}^\perp = \left[-\frac{\partial}{\partial y} \quad \frac{\partial}{\partial x}\right]^\top$, the free-surface multilayer QG equations (in the absence of external forcing) in a layer i read

$$q_i = \Delta\,\psi_i - f_0^2 (\mathsf{A}\underline{\psi})_i, \tag{1a}$$

$$\frac{\partial q_i}{\partial t} + \boldsymbol{\nabla}^\perp \psi_i \cdot \boldsymbol{\nabla}\, q_i = 0, \tag{1b}$$

where the stretching matrix A, is given by

$$\mathsf{A} = \begin{bmatrix} \frac{1}{H_1 g_1} + \frac{1}{H_1 g_2} & -\frac{1}{H_1 g_2} & \cdot & \cdot \\ -\frac{1}{H_2 g_2} & \frac{1}{H_2 g_2} + \frac{1}{H_2 g_3} & -\frac{1}{H_2 g_3} & \cdot \\ \vdots & \ddots & \ddots & \vdots \\ \cdot & \cdot & -\frac{1}{H_N g_N} & \frac{1}{H_N g_N} \end{bmatrix}. \tag{2}$$

2.2 The reduced-gravity framework

Direct observations through satellite sensing are only able to measure sea surface height (SSH), which can be matched to top layer stream function ψ_1. Remote sensing alone can not provide direct measurement on the sublayer flow ψ_2. However, the top layer PV (q_1), the dynamics of which govern surface flow, depends on the stream function from layers 1 and 2 as

$$q_1 = \Delta\,\psi_1 - \frac{f_0^2}{H_1 g_1}\psi_1 - \frac{f_0^2}{H_1 g_2}\left(\psi_1 - \psi_2\right). \tag{3}$$

To overcome such limitation, the RG framework considers the sublayer to be at rest, hence $\psi_2 = 0$. The corresponding top-layer PV q^{RG} expression yields

$$q^{RG} = \Delta\,\psi_1 - f_0^2\left(\frac{1}{H_1 g_1} + \frac{1}{H_1 g_2}\right)\psi_1, \tag{4}$$

which is advected by ψ_1, hence,

$$\frac{\partial q^{RG}}{\partial t} + \boldsymbol{\nabla}^{\perp}\psi_1 \cdot \boldsymbol{\nabla} q^{RG} = 0. \tag{5}$$

This is the formulation of the RG model which was used in the first time to perform dynamical interpolation (Ubelmann et al., 2015, 2016). Benefiting from the ability of the QG model to generate non-linear dynamics, such method has been able to outperform statistics-based interpolation. However, due to the truncated QG equations, significant errors can emerge when performing optimal interpolation using solely the RG model, as pointed out in Manucharyan et al., 2021. In order to address this limitation, the VarDyn framework (Le Guillou et al., 2023, 2025) acknowledges the model error by introducing an external forcing, ϵ, into the advection equation,

$$\frac{\partial q^{RG}}{\partial t} + J(\psi_1, q^{RG}) = \epsilon. \tag{6}$$

This external forcing may mitigate any source of error, not only the truncation error, but also ageostrophic contribution for instance or any imprecision in the model parameters.

2.3 Expressing the missing term from the RG equations

Given the QG PV definition, the top layer PV can be written

$$q_1 = q^{RG} + \frac{f_0^2}{H_1 g_2}\psi_2. \tag{7}$$

Using this decomposition into the top layer PV advection equation yields

$$\frac{\partial q^{RG}}{\partial t} + J(\psi_1, q^{RG}) = -\frac{f_0^2}{H_1 g_2}\left[\frac{\partial \psi_2}{\partial t} + \boldsymbol{\nabla}^{\perp}\psi_1 \cdot \boldsymbol{\nabla}\psi_2\right] \tag{8a}$$

$$= -\frac{f_0^2}{H_1 g_2}\frac{\mathrm{D}\psi_2}{\mathrm{D}t}. \tag{8b}$$

From this expression, we note that the usual RG model advection matches the multilayer formulation when $\mathrm{D}_t\psi_2 = 0$. Addressing the truncation error of the RG model resumes to emulating the right-hand side of eq. (8). Since, ψ_1 is the prognostic variable of the model, computing $\mathrm{D}_t\psi_2$ boils down to estimating ψ_2. Denoting $\tilde{\psi}_2$ such estimation, we define the corrected RG advection by

$$\frac{\partial q^{RG}}{\partial t} + J(\psi_1, q^{RG}) = -\frac{f_0^2}{H_1 g_2}\frac{\mathrm{D}\tilde{\psi}_2}{\mathrm{D}t}. \tag{9}$$

In the following, this model will be referred to as RG-SI (Reduced Gravity - Sublayer Inferred). Additionally to explicitly addressing the RG-related truncation error, this method also provides a proxy for the sublayer stream function, through $\tilde{\psi}_2$. Our specific

parameterization of $\tilde{\psi}_2$ will be detailed in section 4. In the forthcoming, all numerical experiments were conducted using a Pytorch implementation of the multilayer QG equations derived from Thiry et al., 2024 (extended to withstand non-zero boundary conditions)[1].

3 Variational assimilation

Provided a set of observations of the surface $(\mathbf{y}_i)_{i\in\{1,\dots,N\}}$, evaluated at times $(t_i)_{i\in\{1,\dots,N\}}$, we estimate the state $(\mathcal{M}(\theta \mid t_i))_{i\in\{1,\dots,N\}}$ using a model $\mathcal{M}$ (RG-SI for example), which depends on the control vector θ. The latter may include any parameter the propagator dynamics depends on (initial and boundary conditions, model parameters...). Estimated states are then processed through the observation operator $\mathcal{H}$ in order to be compared to the observations. Finally, variational assimilation reduces to finding

$$\theta^\star = \arg\min_\theta \sum_{i=1}^{N} \|\mathbf{y}_i - \mathcal{H}\circ\mathcal{M}(\theta \mid t_i)\|^2_{\mathsf{R}^{-1}} = \arg\min_\theta \mathcal{J}_{\text{obs}}(\theta)\,, \tag{10}$$

with the observation error covariance matrix R.

However, there are no guarantees that enough data is available for the minimization problem to be well constrained. Therefore, provided a prior estimate of the control vector, θ^b, and its error covariance matrix, B, the cost function is usually regularized by penalizing the deviation of θ from θ^b. Introducing the regularization loss

$$\mathcal{J}_{\text{reg}}(\theta) = \|\theta - \theta^b\|^2_{\mathsf{B}^{-1}}, \tag{11}$$

the cost function rewrites

$$\mathcal{J}(\theta) = \mathcal{J}_{\text{obs}}(\theta) + \mathcal{J}_{\text{reg}}(\theta)\,. \tag{12}$$

Supposing we can estimate the gradient $\boldsymbol{\nabla}\mathcal{J}$ of the loss function, minimization can be performed through gradient descent. Regarding our specific experimentations, since all models and operation are implemented using Pytorch, we benefit from the library's built-in automatic differentiation (Paszke et al., 2017) to both retrieve loss function gradient and perform a descent algorithm.

4 The forward model

4.1 Sublayer parameterization

In order to keep a reasonable number of parameters for the parameterization of $\tilde{\psi}_2$, we use a reduced-order space to represent its value. We have decided to express $\tilde{\psi}_2$ over a quite simple space so that RG-SI performances could not be attributed to the structure of the parameterization rather than the actual error model. Therefore, the spatio-temporal behavior of $\tilde{\psi}_2$ is prescribed through a partition of unity of Gaussian envelopes, both in

[1] Code available at https://github.com/grigaut/QG/

time and space. We denote N_T the number of time-related envelopes and N_{XY} the number of space-related envelopes at a given time t_i. Given $i \in \{1, \dots, N_T\}$ and $j \in \{1, \dots, N_{XY}\}$, we define

$$e_i^T(t) = \exp\left[-\left(\frac{t - t_i}{\sigma_T}\right)^2\right], \tag{13a}$$

$$e_j^{XY}(x, y) = \exp\left[-\left(\frac{x - x_j}{\sigma_X}\right)^2\right] \exp\left[-\left(\frac{y - y_j}{\sigma_Y}\right)^2\right], \tag{13b}$$

where t_i, x_j, y_j are the locations of the envelope centers and $\sigma_T, \sigma_X, \sigma_Y$ their standard deviation. Envelopes centers are chosen so that they are evenly spaced (whether in time or space). Details on the determination of the standard deviations can be found in appendix A.

We denote $E^T = \sum_{i=1}^{N_T} e_i^T$ and $E^{XY} = \sum_{i=1}^{N_{XY}} e_j^{XY}$. Using these notations, the overall decomposition of $\tilde{\psi}_2$ at time t and location (x, y) reads

$$\tilde{\psi}_2(t, x, y) = \frac{\sum_{i=1}^{N_T} \sum_{j=1}^{N_{XY}} \theta_{i,j} e_i^T(t)\, e_j^{XY}(x, y)}{E^T(t)\, E^{XY}(x, y)}, \tag{14}$$

where the $(\theta_{i,j}) \in \mathbb{R}^{N_T \times N_{XY}}$ are coefficients inferred through cost function minimization. Such parameterization allows representing a smooth field while enabling the emergence of some rather small scale patterns. Additionally, we can easily restore the dynamics of a flow under the RG assumption by setting all coefficients to 0. In the forthcoming, still will be the initial condition considered when instantiating the minimization of the cost function.

4.2 Physics-based regularization

Following eq. (1b), PV advection in the layer 2 reads

$$\partial_t q_2 + \boldsymbol{\nabla}^\perp \psi_2 \cdot \boldsymbol{\nabla} q_2 = 0. \tag{15}$$

However, as the shape of our model error yields a proxy for ψ_2, we define the sublayer PV using the truncation of its multilayer definition (we consider $\psi_3 = 0$), retrieving the following proxy for q_2,

$$\tilde{q}_2 = \Delta \tilde{\psi}_2 - \frac{f_0^2}{H_2 g_2}\left(\tilde{\psi}_2 - \psi_1\right). \tag{16}$$

We can compute an estimation of the sublayer advection residual with $\partial_t \tilde{q}_2 + \boldsymbol{\nabla}^\perp \tilde{\psi}_2 \cdot \boldsymbol{\nabla} \tilde{q}_2$. In order to enforce physical constraints during the optimization process, we introduce a physics-based regularization,

$$\mathcal{J}_{\text{reg}}\left(\psi_1, \tilde{\psi}_2\right) = \left\|\partial_t \tilde{q}_2 + \boldsymbol{\nabla}^\perp \tilde{\psi}_2 \cdot \boldsymbol{\nabla} \tilde{q}_2\right\|^2, \tag{17}$$

and we define the overall cost function with

$$\mathcal{J}\left(\tilde{\psi}_2\right) = \mathcal{J}_{\text{obs}}\left(\tilde{\psi}_2\right) + \gamma \mathcal{J}_{\text{reg}}\left(\tilde{\psi}_2\right), \tag{18}$$

where γ is an hyperparameter driving the relative importance of the regularization over the observation loss. Replacing the usual background state related regularization by this physics-based formulation is inspired by physics-informed neural networks (PINN) methodology (Raissi et al., 2019) and PINN-derived data assimilation approaches have proven the relevancy of regularizing using physical constraint (Oh, 2025; Ugur & Zhou, 2025).

Additionally, switching to a physics-based regularization removes the need for an a priori guess of the parameter. Since our method is based on reconstructing a proxy for the sublayer stream function, retrieving a non-trivial estimation for ψ_2 remains challenging. A possible estimation would have been to penalize $\tilde{\psi}_2$ from deviating from 0, thus preventing the model from stirring away from the RG framework. It would have also been possible to enforce a barotropic flow by penalizing deviation from ψ_1. However, both of these options pull the model toward one-layer dynamics whereas the physics-derived regularization allows for two-layer-like dynamics while still allowing the system to fall back to $\tilde{\psi}_2 \simeq 0$ or $\tilde{\psi}_2 \simeq \psi_1$ when appropriate.

It is also worth emphasizing that introducing this regularization does not strictly enforce the conservation of PV in the layer. It guides the model towards finding an expression for $\tilde{\psi}_2$ that approaches the conservation $\tilde{q}_2$. However, as there are no guarantees that $\tilde{q}_2$ should be strictly conserved ($\tilde{\psi}_2$ is not exact and $\tilde{q}_2$ is computed through the truncation of the multilayer equations), this formulation allows for some flexibility on said conservation.

Furthermore, if we were to assimilate an initial condition ψ_2^0 for the bottom layer stream function in two-layer QG model, constraining the minimization problem would yield the question of finding a background estimation for ψ_2^0. Once again, two natural choices would be the following: considering 0 or ψ_1^0 (or a linear combination of both, as done by Cong and Ikeda, 1995), hence leaning toward either a surface intensified or a barotropic initial condition for the flow. As the background term would strongly call back the ψ_2^0 toward one of these options, the overall dynamics would be notably impacted by this decision. Indeed, according to Hurlburt, 1986, instantiating a two-layer model by setting the initial sublayer field at rest yields not better results regarding surface forecast than a regular RG model, thus emphasizing the need for a precise a priori estimation of sublayer initial state. Our regularization however, does not constrain the value of the sublayer stream function itself but rather its dynamics: it has to somehow satisfy a pseudo-advection of PV in the bottom layer, but as for its value it remains unconstrained. Additionally, this method allows for more flexibility regarding subsurface flow: if improving surface dynamics requires drifting away from the advection of the sublayer (truncated) PV (through the introduction of non-QG patterns in the reconstructed field for example), it can do so. A strictly enforced two-layer QG model would not be able to do so without the introduction of an external forcing term in the sublayer advection equation.

4.3 Deformation radius correction

Other than the truncation error, incorrect model stratification parameters (in the H_i or the g_i) could result in mismatch in the surface flow dynamics. Key characteristics of the flow dynamics are the so-called barotropic and first baroclinic deformation radii, which, in the multilayer QG framework, can be computed from the eigenvalues of the stretching matrix A. In order to increase the robustness of our two-layer-like model, we wish to allow for baroclinic deformation radius adjustments. Given the diagonalization of the two-layer stretching matrix,

$$\mathsf{A} = \mathsf{C}_{\mathrm{M2L}} \begin{bmatrix} \lambda_{\rightleftarrows} & 0 \\ 0 & \lambda_{\rightrightarrows} \end{bmatrix} \mathsf{C}_{\mathrm{L2M}}, \tag{19}$$

where $\lambda_{\rightleftarrows}$ is the baroclinic-related eigenvalue and $\lambda_{\rightrightarrows}$ the barotropic one, and $\mathsf{C}_{\mathrm{M2L}}$ (resp., $\mathsf{C}_{\mathrm{L2M}}$) the modes to layers (resp., layers to modes) passage matrix. The baroclinic radius correction is performed by introducing the parameter α, and the modified stretching matrix,

$$\tilde{\mathsf{A}}_\alpha = \mathsf{C}_{\mathrm{M2L}} \begin{bmatrix} (1+\alpha)\,\lambda_{\rightleftarrows} & 0 \\ 0 & \lambda_{\rightrightarrows} \end{bmatrix} \mathsf{C}_{\mathrm{L2M}}, \tag{20a}$$

$$= \mathsf{A} + \alpha\lambda_{\rightleftarrows} \mathsf{C}_{\mathrm{M2L}} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathsf{C}_{\mathrm{L2M}}, \tag{20b}$$

which coincides with the default stretching matrix when $\alpha = 0$. The barotropic deformation radius then remains unchanged as it is linked to the total depth which can be quite easily approximated whereas the baroclinic deformation radius, $L_{\rightleftarrows}$ becomes

$$L_{\rightleftarrows} = \frac{1}{f_0\sqrt{(1+\alpha)\,\lambda_{\rightleftarrows}}}. \tag{21}$$

Our model implementation replaces A by $\tilde{\mathsf{A}}_\alpha$ in the usual two-layer QG equations and the parameter α is added in the control vector, granting one additional degree of freedom for the variational assimilation.

5 Method validation

We have considered two validation experiments. In the first one, we have set a double-gyre experiment using a three-layer QG model (QG3L). Therefore, the reconstruction error from the RG model (dynamical interpolation situation) is mostly explained by the truncation of the QG equations. Within this framework, we can also assess the ability of RG-SI model in reconstructing top layer dynamics and compare its performances with a RG model with no error correction (RG0) and a RG model whose error is prescribed though an external forcing, RG-F (RG-Forced). Two variants of this setup have been explored. One where model parameters (reference layer depth, reduced gravity values…) are the same in the reference QG model and in the RG counterpart. This configuration confines the error

sources to truncation error while also allowing for a fair comparison of $\tilde{\psi}_2$ and ψ_2. In the other variant, model parameters are modified to assess the robustness of each of the models when dealing with incorrect baroclinic deformation radius. The second validation experiment uses surface fields from the eNATL60 simulation (Brodeau et al., 2020) as the reference flow. This experiment investigate the capability of our model in reproducing a flow that present ageostrophic, complex and rich dynamical features.

As this work focuses on introducing a new parameterization to reduce model error, we have decided to only consider such parameterization in the control vector. Indeed, since all tested frameworks (RG0, RG-F, RG-SI) share identical initial and boundary conditions, differences in performances only results from model error and its correction. Ideal initial and boundary conditions are therefore retrieved from the surface flow of the reference simulation. Furthermore, the estimation of initial and boundary conditions through variational assimilation has already been widely investigated (Ayoub, 2006; Cong et al., 1998; Gunson & Malanotte-Rizzoli, 1996; Seiler, 1993; Zou et al., 1995), suggesting that our methodology could quite easily be adapted to a more realistic data assimilation configuration.

In both experiments, observations are drawn from the surface using an observing system simulation experiment that simulates SWOT-like sampling tracks. Satellite track example can be found in fig. 1, along with the cumulative passages of the tracks over a 20-days period.

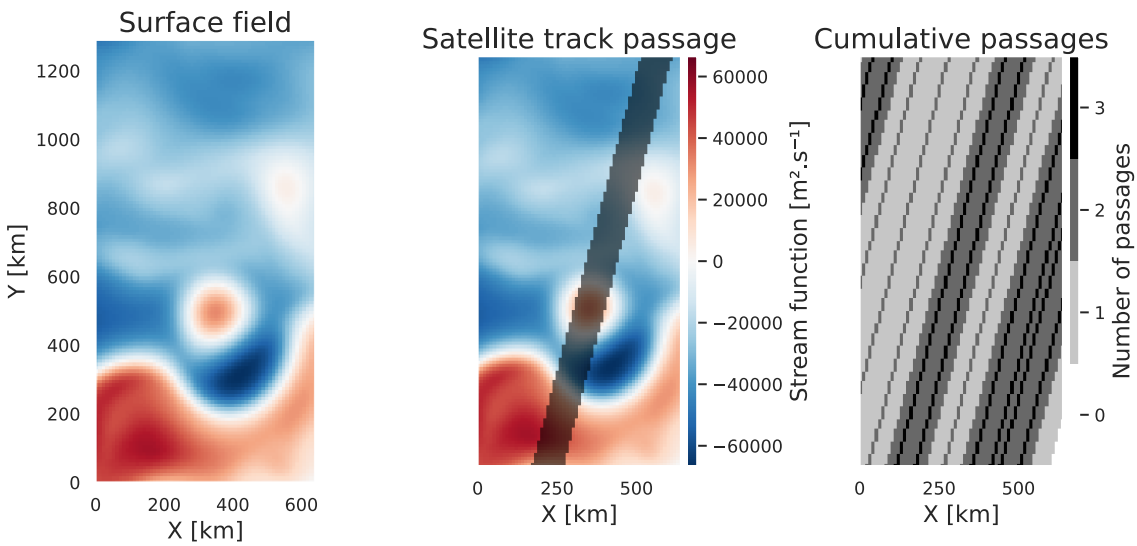

Figure 1: Example of simulated satellite track over the surface (Surface fields extracted from a QG3L double-gyre simulation). Last panel shows how many times has a pixel been observed over the 20-days period.

Assimilation is conducted on 20 days-long periods. Over these periods, satellite passages are simulated and, when a track spans over the considered domain, observation loss is evaluated using the collected data. Regarding the RG-SI model, regularization loss is evaluated at every time step. Details on the implementation of RG-F forcing and regularization are detailed in appendix B.

During the variational assimilation process, all parameterized field are instantiated at

0. Therefore, RG-SI's initial observation loss is the exact same as the RG one since the bottom layer appears to be at rest. Loss minimization is performed using Pytorch's implementation of the Adam optimizer (Kingma & Ba, 2017). Optimization was stopped after 200 iterations as longer optimization did not yield any significant improvement in the loss function for each of the models.

After the assimilation process is completed, model performances are evaluated using the normalized Root Mean Squared Error (nRMSE). Given a field χ and its reference value χ_{ref}, the corresponding nRMSE is

$$\text{nRMSE}\left(\chi, \chi_{\text{ref}}\right) = \sqrt{\frac{\left\|\chi - \chi_{\text{ref}}\right\|^2}{\left\|\chi_{\text{ref}}\right\|^2}}. \tag{22}$$

Therefore, perfect field reconstruction would result in a nRMSE score of 0 whereas a nRMSE score above 1 indicates that no useful information on χ_{ref} can be retrieved from χ.

5.1 Multilayer QG

5.1.1 Experimental set-up

Models parameters were set so that all models would share same time-space resolutions and evolve on the same β-plane. Regarding stratification parameters, RG0 has been implemented as a one-layer QG model with $g = g_1 g_2/(g1 + g_2)$ so that it matches the parameters of a truncation of the multilayer equations (see eq. (4)). RG-SI and RG-F have been instantiated so that they share the same deformation radius, namely 29 km, which corresponds to the baroclinic deformation radius of the two-layer QG model matching the parameters of the reference three-layer QG on the top two-layers. Such deformation radius value is also similar to the one described in Le Guillou et al., 2023. All model parameters are detailed in the following table

Parameter	Value			
$\mathrm{d}t$	7200 s			
$\mathrm{d}x$	10 km			
$\mathrm{d}y$	10 km			
f_0	$9.375 \times 10^{-5}\,\mathrm{s}^{-1}$			
β	$1.754 \times 10^{-11}\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$			
Parameter	QG3L	RG0	RG-F	RG-SI
H_1	400 m	400 m	293 m	400 m
H_2	1100 m	.	.	1100 m
H_3	2600 m	.	.	.
g_1	$9.81\,\mathrm{m/s^2}$	$0.0249\,\mathrm{m/s^2}$	$0.025\,\mathrm{m/s^2}$	$9.81\,\mathrm{m/s^2}$
g_2	$0.025\,\mathrm{m/s^2}$	.	.	$0.025\,\mathrm{m/s^2}$
g_3	$0.0125\,\mathrm{m/s^2}$	.	.	.
R_d	41 km	34 km	29 km	29 km

Table 1: Model parameters. R_d denotes the baroclinic deformation radius.

Since the reference simulation is forced by wind to display a double-gyre, we also include the wind as an external forcing within the advection equations for all models, denoting $\tau = \begin{bmatrix}\tau_x & \tau_y\end{bmatrix}^\top$ the wind stress tensor, the advection equation rewrites

$$\frac{\partial q^{RG}}{\partial t} + \boldsymbol{\nabla}^\perp \psi_1 \cdot \boldsymbol{\nabla}\, q^{RG} = F + \boldsymbol{\nabla}^\perp \cdot \tau \tag{23}$$

where F represent the additional forcing that depends on the frameworks,

$$F_{\mathrm{RG}} = 0, \tag{24a}$$
$$F_{\mathrm{RG-F}} = \epsilon, \tag{24b}$$
$$F_{\mathrm{RG-SI}} = \frac{-f_0^2}{H_1 g_2}\frac{\mathrm{D}\tilde{\psi}_2}{\mathrm{D}t}. \tag{24c}$$

As the reference was established over a 2560 km by 5120 km domain, we have decided to focus on subdomains of the said area. We selected two 640 km x 1280 km zones, Z1, Z2

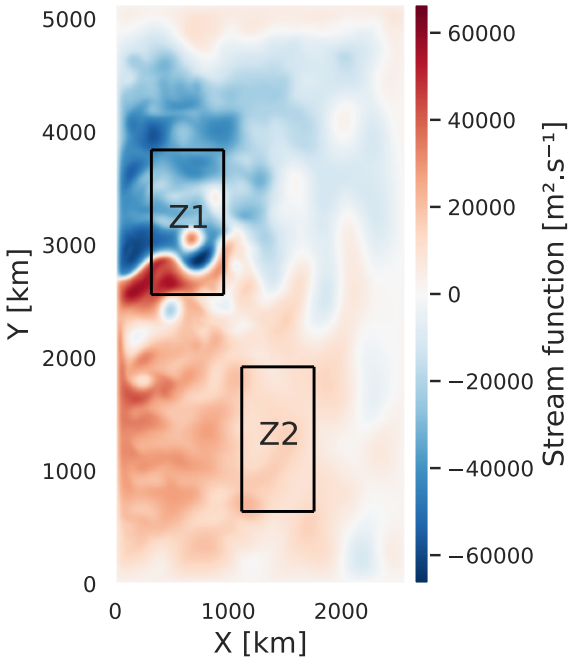

Figure 2: Detail of the 4 zones of the three-layer QG experiment.

We have selected these two zones as they present different dynamical features. Z1 for instance is at the core of the eastward jet and presents a surface intensified dynamics. On Z2 however the baroclinic components of the flow is stronger. In both zones, the OSSE follows the same tracks, observing 13243 gridpoints. Number of control parameters for the different models are detailed in the following table

	RG-SI	RG-SI ($\alpha = 0$)	RG-F	RG0
Control vector size	6322	6321	8512	0

Table 2: Comparison of the control vector size for the different models in the QG3L experiment.

Both RG-F and RG-SI regularizations are scaled through an hyperparameter (γ). The value of γ has been taken constant over all areas and assimilation periods. Regarding RG-SI, its value has been fitted so that regularization loss on first assimilation step was around 1%-1‰ of the observation loss in all areas and periods. Regarding RG-F, the value γ has been empirically chosen so that the model would perform as best as possible while not providing field that were small-scale intensified (which it tends to do in the absence of regularization). However, across all experimentations (either regarding RG-SI or RG-F), quality of the reconstruction showed small sensitivity to the value of γ for a given order of magnitude.

5.1.2 Surface stream function reconstruction evaluation

Results (averaged on 3 consecutive 20-days periods) are shown in fig. 3. For each zone, we perform variational assimilation on 12 20-days periods, each separated by 20 days. The resulting nRMSEs are averaged over these 12 periods and the nRMSE standard deviation is computed to display the variability of the scores across the different periods.

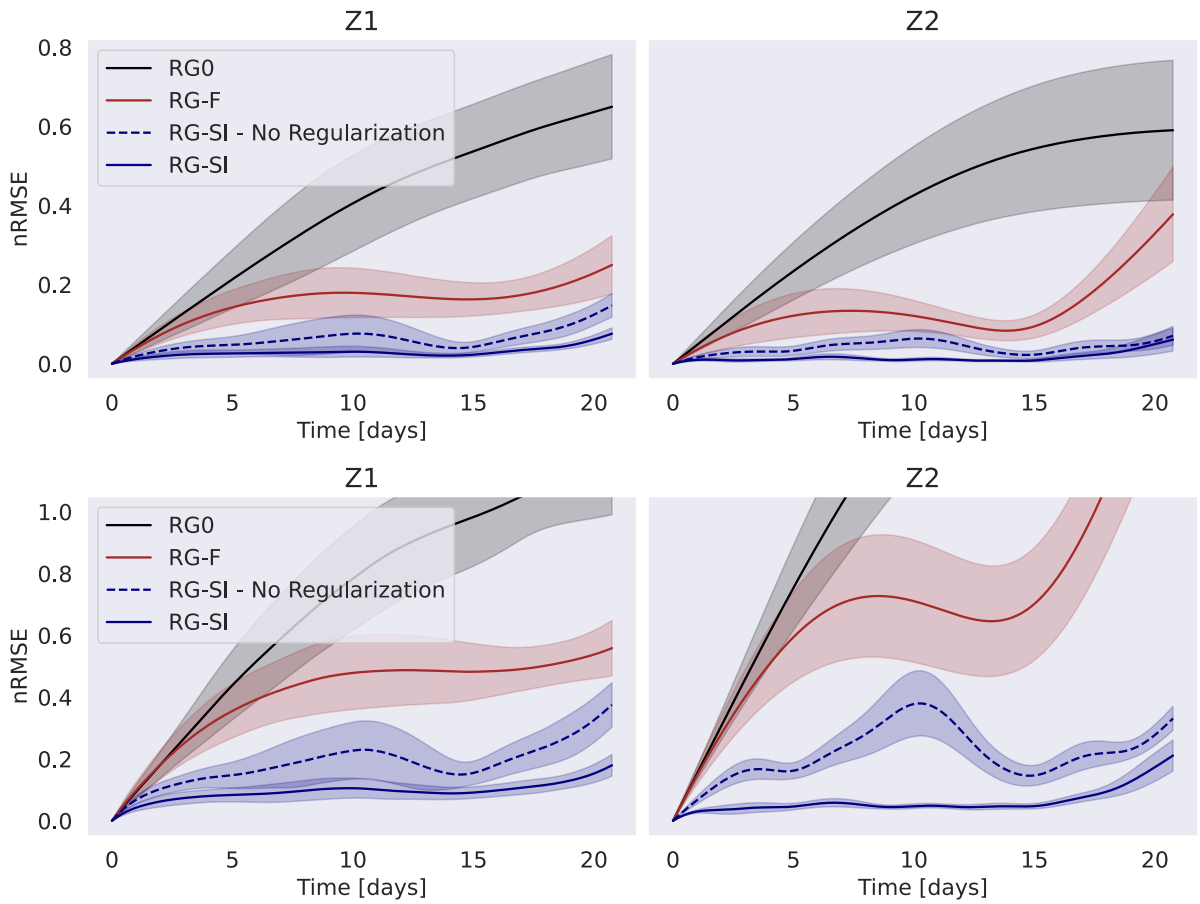

Figure 3: nRMSE on three-layer QG surface stream function (top) and velocities (bottom) (averaged over 12 20-days periods, each separated by 20 days) on Z1 (left) and Z2 (right). Lines represent the nRMSE average and the shaded area the standard deviation. Models are identified by the line color, black for RG, brown for RG-F and blue for RG-SI. The dashed line represents the non-regularized RG-SI (hence $\gamma = 0$ in eq. (18)). Y-axis is shared by both panels.

The RG-SI method is able to reconstruct the surface state with high fidelity (averaged nRMSE on Z1 is about 0.029 and about 0.016 on Z2 as detailed in table 3). The introduction of the regularization, not only allowing the variational problem to be well constrained, also halves the average nRMSE over both zones. It also significantly reduces the empirical standard deviation, suggesting that the regularization helps the model in consistently converging toward a better reconstructed field.

	RG-SI	RG-SI (No Reg)	RG-F	RG0
Z1	0.029	0.060	0.15	0.38
Z2	0.016	0.040	0.13	0.38

Table 3: Averaged nRMSE values.

Comparison with other methods shows that RG-SI outperforms both RG0 and RG-F. Such result highlights that the introduction of an external (the RG-F methodology), although

able to mitigate some of the RG-truncation error, is not as effective as the truncation-aware method as introduced in RG-SI. It is also worth mentioning that, as a comparison, we have tried fitting $\tilde{\psi}_2$ using direct (full-field) observation from the ψ_2 field of the QG3L model. Plugging the resulting parameters into the RG-SI model yields similar results as RG-SI (corresponding figure can be found in the appendix fig. C.1, averaged nRMSE values are of 0.041 in Z1 and 0.012 in Z2).

The visual comparison of vorticity fields shows almost identical fields. An example of such fields after 5 and 20 days of simulation are displayed in fig. 4

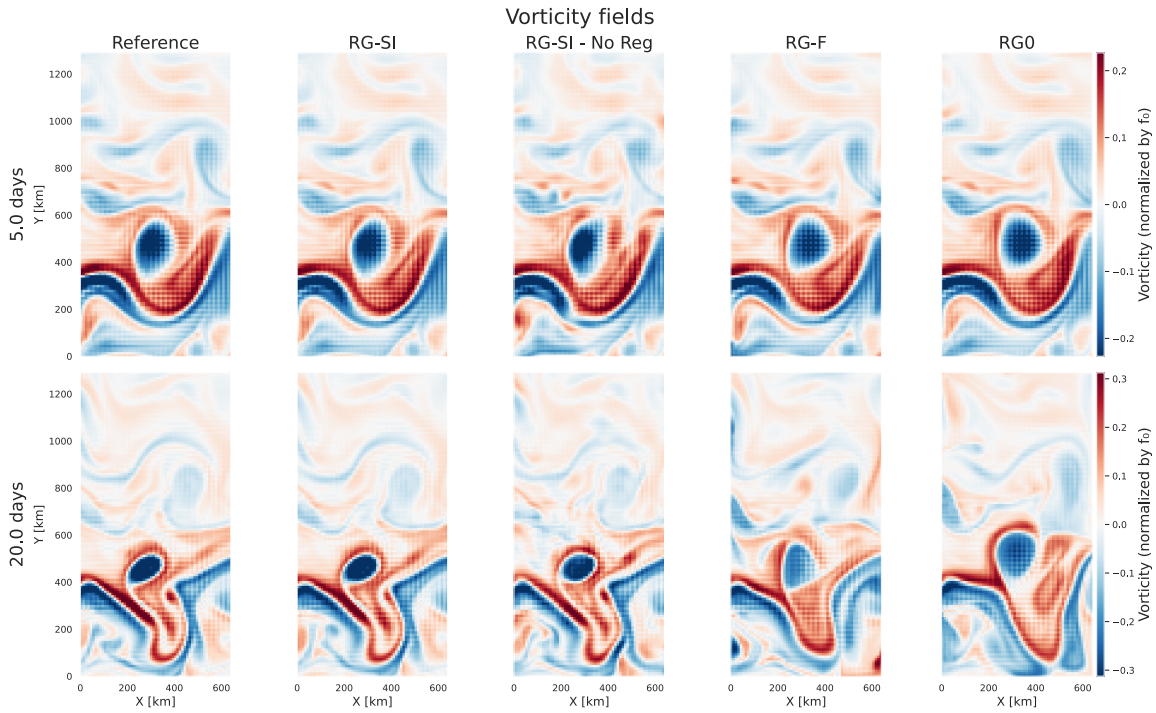

Figure 4: Surface vorticity fields retrieved from first assimilation cycle performed in Z1, after variational minimization has been completed. First column displays Reference vorticity field while second, third, fourth and fifth show RG-SI, non-regularized RG-SI, RG-F and RG0 reconstructions, respectively. Top row displays vorticity fields after 5 days and bottom one after 20 days.

A careful comparison of regularized (second column) and non-regularized (third column) vorticity fields obtained with the RG-SI framework shows that the regularization improves small scale reconstruction by removing some small scale discrepancies.

5.1.3 Estimation of the sublayer stream function

As the RG-SI method provides an estimation for the sublayer flow, ψ_2, we have evaluated the reconstruction of ψ_2 by $\tilde{\psi}_2$. Given the reference sublayer stream function from the three-layer QG simulation, nRMSE is evaluated for $\tilde{\psi}_2$. Averages and standard deviation are computed over the variational assimilation periods as for the surface stream function.

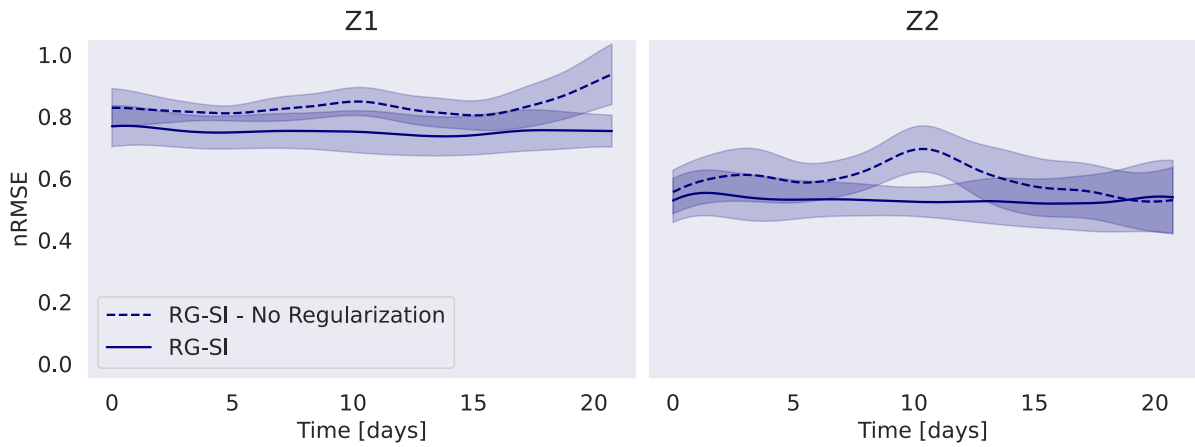

Figure 5: nRMSE on three-layer QG sublayer stream function (averaged over 12 20-days periods, each separated by 20 days) on Z1 (left) and Z2 (right). RG-SI model is identified by the solid blue line and non-regularized RG-SI by the dashed blue line. Y-axis is shared by both panels.

Although results show that the regularization improves the reconstruction of ψ_2 by $\tilde{\psi}_2$, the quality of the overall reconstruction is far from being as good as it was for the surface flow. However, although the parameterized field is indeed ψ_2, the actual field is never directly used by the model. What matters when emulating the dynamics is $\mathrm{D}_t\tilde{\psi}_2$. Therefore, although ψ_2 might not be accurately represented by $\tilde{\psi}_2$, as long as $\mathrm{D}_t\psi_2$ is by $\mathrm{D}_t\tilde{\psi}_2$, the surface dynamics can still be previsely reconstructed. Evaluating the nRMSE of $\mathrm{D}_t\tilde{\psi}_2$ yields better results, as it is mostly stable around 0.5 (resp. 0.15) in Z1 (resp. Z2). Corresponding figures can be found in the appendix (fig. C.2). Although still not as good as top layer reconstruction scores, sublayer stream function material time derivative is better reconstructed than the field itself. Actually, when projecting the sublayer stream function onto the reduced-order space, the resulting error in the material derivative of the sublayer stream function is not very far from the one obtained using RG-SI. Once again, results show the benefit of using the physics-based regularization.

5.1.4 Robustness to stratification parameters perturbations

To assess the relevancy of the deformation radius correction presented in section 4.3, we keep the same parameters for the three-layer QG reference model, and we modify the other models' ones. Stratification perturbation are prescribed by modifying the layer reference thickness of the models. New parameters and the resulting deformation radius are detailed in the following table

Parameter	QG3L	RG	RG-F	RG-SI
H_1	400 m	150 m	135 m	150 m
H_2	1100 m	.	.	1350 m
H_3	2600 m	.	.	.
R_d	41 km	21 km	19 km	19 km

Table 4: Model parameters in the perturbed framework. Other model parameters were retained as shown in table 1

We then conducted the same experiment as before and monitored the ability of the different models to tamper incorrect parameters and reconstruct the surface dynamics.

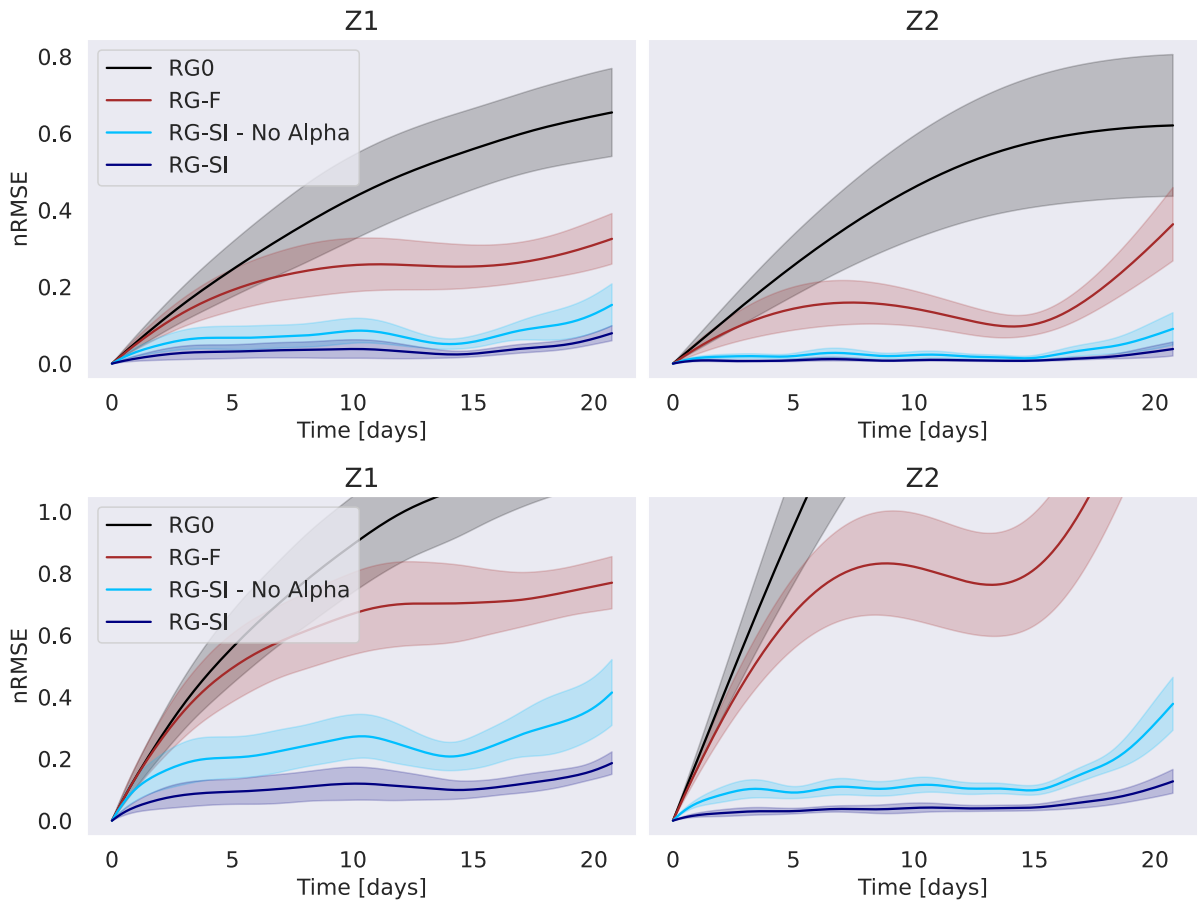

Figure 6: nRMSE on three-layer QG surface stream function and velocities (average over 12 different assimilation periods each of 20 days) on Z1 (left) and Z2 (right) using perturbed models. Lines represent the nRMSE average and the shaded area the standard deviation. Models are identified by the line color, black for RG, brown for RG-F and dark blue for RG-SI. A light blue line is used for the RG-SI model with $\alpha = 0$ (hence no deformation radius correction).

Here again, RG-SI accurately reconstructs the surface flow dynamics. The comparison of error scores of the standard RG-SI model and its counterpart with fixed $\alpha = 0$ shows that the introduction of a single additional parameter into the control vector results in an error reduction of more than 2 (see table 5).

	RG-SI	RG-SI ($\alpha = 0$)	RG-F	RG0
Z1	0.034	0.072	0.22	0.40
Z2	0.011	0.027	0.14	0.41

Table 5: Averaged nRMSE values.

Relative comparison with the error scores from table 3 yet show that, overall, RG-SI still performs worse in Z1 in the perturbed framework ($\sim 17\%$ relative error degradation) and,

surprisingly performs better in Z2 ($\sim 31\%$ improvement). RG-F performances, however, are both degraded in the perturbed framework, notably in Z1 ($\sim 47\%$ relative degradation) and slightly in Z2 ($\sim 8\%$ degradation), although slightly in Z2. Nonetheless, perturbed RG-SI still outperform non-perturbed RG-F in reconstructing the surface flow of the three-layer QG model.

5.2 Validation with realistic OSSE

Using surface fields from the eNATL60 simulation (Brodeau et al., 2020), we monitor RG-SI's reconstruction of a fluid with ageostrophic dynamics. We have focused on a single area of dimension 680 km x 1030 km centered on 55°W 40°N. Parameters for all models are detailed are the exact same as the ones from table 1, apart from Coriolis parameters which were set to $f_0 = 9.373\,\mathrm{s}^{-1}$ and $\beta = 1.753\,\mathrm{m}^{-1}\,\mathrm{s}^{-1}$. Surface stream function anomaly is inferred from eNATL60's SSH and atmospheric pressures anomalies, the latter being obtained from the Drakkar Forcing Set 5.2 (Dussin et al., 2016). Anomalies are computed by removing its space-average from the given field. Therefore, introducing the space-average operator $\langle SSH \rangle_{XY}$, surface stream function anomaly Ψ_{surf} reads

$$\Psi_{\mathrm{surf}} = \frac{p_{atm}}{f_0 \rho_0} + \frac{g}{f_0} SSH - \left(\frac{\langle p_{\mathrm{atm}} \rangle_{XY}}{f_0 \rho_0} + \frac{g}{f_0} \langle SSH \rangle_{XY} \right), \tag{25}$$

where ρ_0 is the reference density for water. Additional experiments were conducted by removing the time-space average of both SSH and atmospheric pressure fields and yielded overall similar results. However, in such configurations, the reference fields contained large-scale spatial fluctuations corresponding to the ocean response to atmospheric pressure. Since the QG models are not able to capture such fluctuations, we decided to mitigate them by removing the space-averaged (instead of the time-space average) stream function components, as illustrated by eq. (25).

As the resulting stream function contain small-scale features, it is then smoothed using a Gaussian filter and then downsampled from the high-resolution, curvilinear coordinates of the eNATL60 simulation into the regular, Cartesian grid of the QG models. It is this smoothed, down-scaled field that is used for both initial and boundary conditions of the QG models. The observations however are sampled from the non-filtered surface stream function field, using the same OSSE as for the three-layer QG comparison. A total of 10333 pixels are observed during a single assimilation cycle. Control vectors size are as follows

	RG-SI	RG-F	RG0
Control vector size	5475	6608	0

Table 6: Comparison of the control vector size for the different models in the eNATL60 experiment.

Once again variational assimilation is conducted over 20-days periods and the quality of the reconstruction is assessed by using nRMSE to compare reference stream function to its reconstructed counterpart. Assimilation is conducted over 4 consecutive periods, during

two different seasons, summer (July, August, September) and winter (January, February, March). Results are shown in fig. 7.

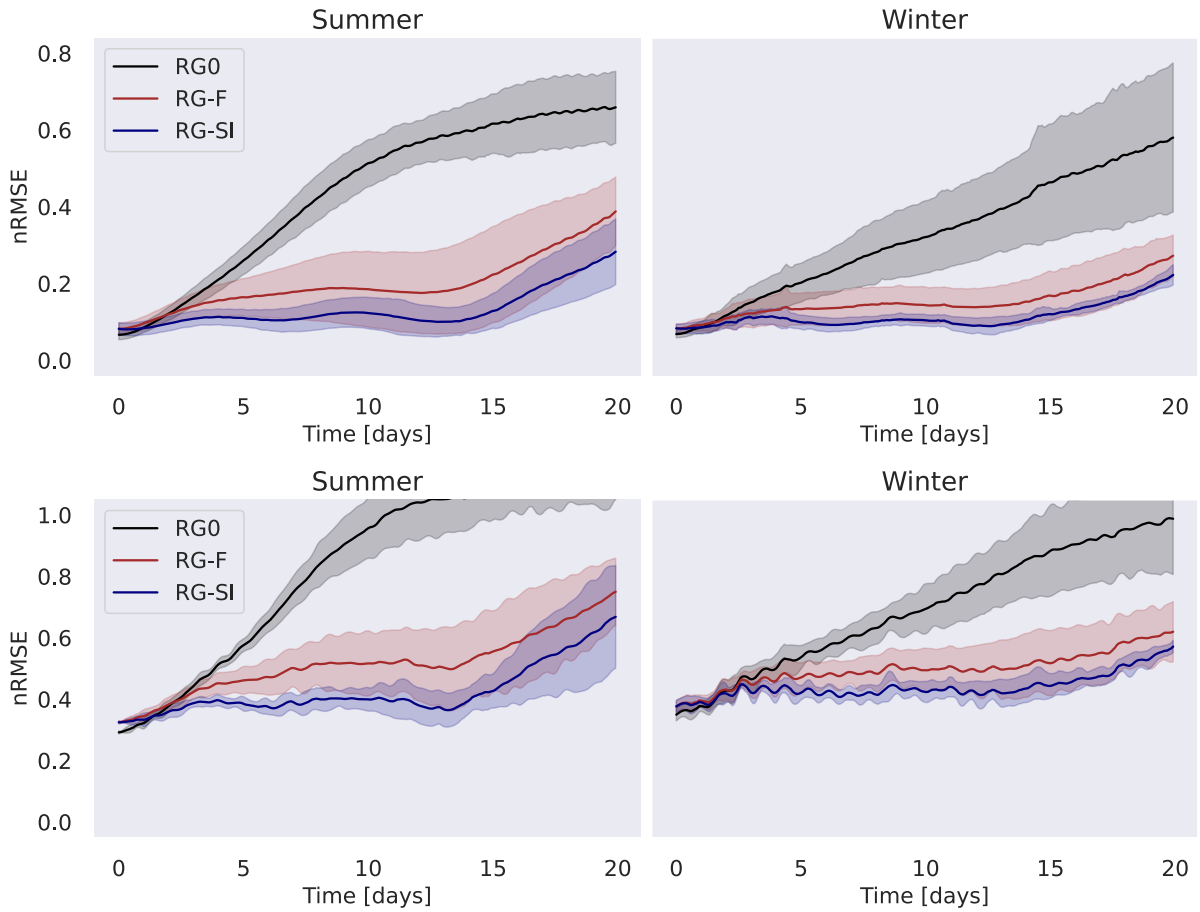

Figure 7: nRMSE on eNATL60 surface stream function and velocities (averaged over 4 consecutive 20-days periods). Summer is displayed in left panel and winter in right one. Models are identified by the line color, black for RG, brown for RG-F and blue for RG-SI. Solid line represents the averaged over the 4 periods and the shaded area the empirical standard deviation. Y-axis is shared by both panels.

	RG-SI	RG-F	RG0
Summer	0.13	0.20	0.44
Winter	0.11	0.15	0.33

Table 7: Averaged nRMSE values.

Results show that RG-SI still outperforms RG-F in reconstructing the top layer flow in a realistic configuration. Such results indicate that a great component of the error conducted by the RG model can be explained by a field advected by geostrophic flow (whether only the sublayer geostrophic stream function or a more complex field). Regardless of the season (summer or winter), RG-SI provides an improvement over the RG-F method and significantly decreases the error that was conducted by the RG0 model.

The visual comparison of the surface vorticity field demonstrates the ability of the RG-SI method in reconstructing a realistic flow. An example of the vorticity fields in shown in fig. 8. The overall comparison of the models show that RG-SI is able to reconstruct larger vortices, although its resolution prevents it from rendering smaller ones. RG-SI also seems to outperform the other models in locating the larger vortices.

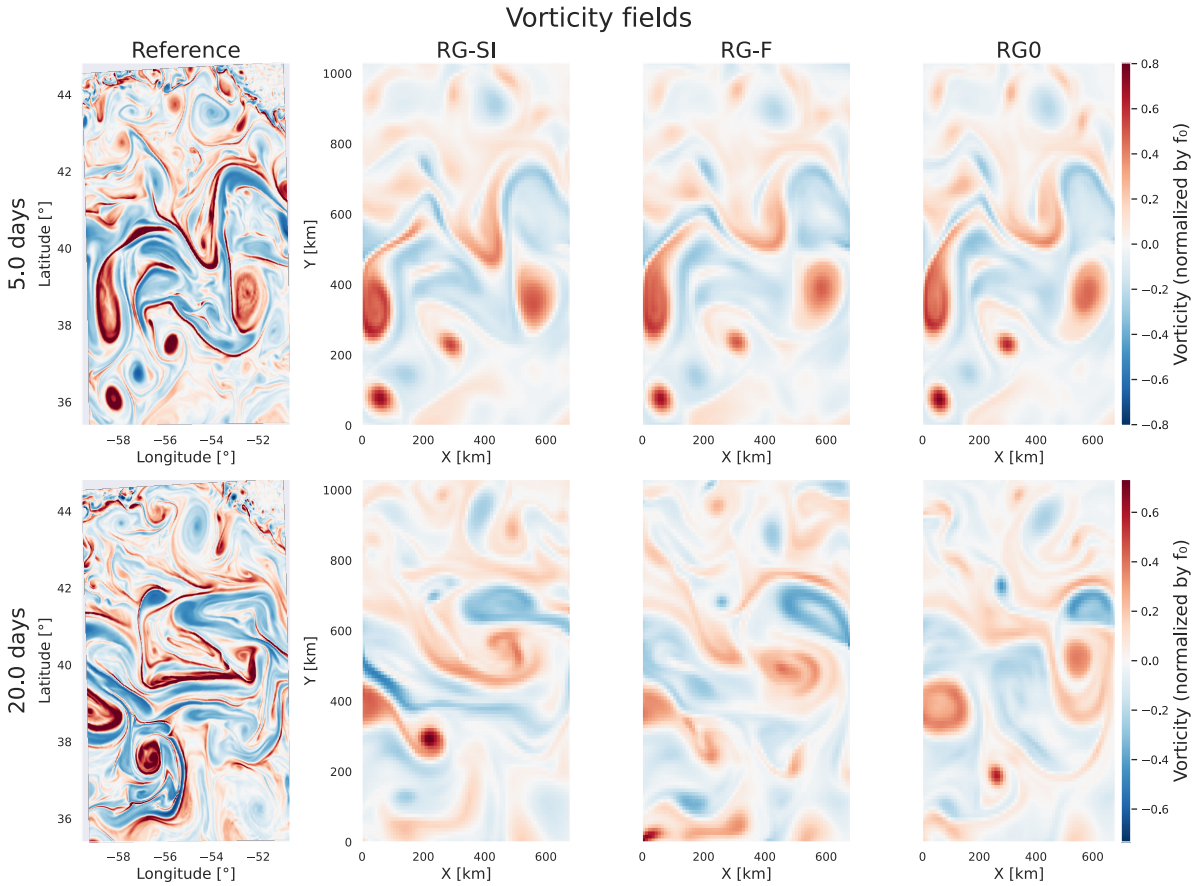

Figure 8: Surface vorticity fields retrieved from first assimilation cycle performed in summer, after variational minimization has been completed. First column displays Reference vorticity field while second, third and fourth show RG-SI, RG-F and RG0 reconstructions, respectively. Top row displays vorticity field after 5 days and bottom one after 20 days.

6 Discussion

In order to improve sea surface reconstruction from altimetric observations using a RG model, we have introduced a new formulation for the model error. This method emulates the sublayer stream function in order to account for the error that inherently ships with the RG assumption. Additionally,

The improvement of surface flow reconstruction achieved by the RG-SI method highlights the benefits of introducing a parameterization for the sublayer stream function. Indeed, such parameterization allows for a significant reduction of the model error induced by the truncation of the multilayer QG equations. The introduction of a physics-based regularization also appears to significantly improve the model performances, as it softly

enforces multilayer-like dynamics while still allowing for some flexibility on the sublayer flow.

Although RG-SI reaches almost perfect surface reconstruction in the idealized three-layer QG framework, its reconstruction of the sublayer stream function is not as accurate. If we were to focus on the sublayer flow reconstruction, we could look for a reduced-order space that encourages physically-relevant patterns for the subsurface stream function. However, enforcing such a QG structure might prevent the model from generalizing to more realistic flows, for which the identification of the parameterized field to the sublayer stream function might not be relevant. Indeed, as the eNATL60 experiment shows, a significant part of the realistic flow is still not captured by the RG-SI method. Rather than enforcing a QG structure through appropriate reduced-order space, improvement might be achieved by digging into the next-order terms of the QG approximation (Muraki et al., 1999). Despite identifying the parameterized field to the sublayer stream function is a natural choice in the multilayer QG framework, such identification can be challenged in a realistic flow. As the QG PV can be obtained through a truncation of the asymptotic decomposition of Ertel's PV (Müller, 1995), the parameterized field might rather be identified to the missing terms of said truncation, or rather to the terms transported by the geostrophic surface flow. Given that the realistic surface contains ageostrophic components, the methodology could be improved by emulating such terms. This could be done through the explicit introduction of additional parameterized fields, or through a combination of the RG-SI and RG-F methodologies, considering that the RG-SI parameterization would tackle the truncation error while leaving to RG-F's forcing all the remaining unresolved processes.

Rather than introducing additional field to the model's control vector, another option would be to integrate external forcing into the method. Bottom topography for example, can have a significant impact on the flow dynamics and can not be taken into account by the RG framework (as pointed out by Hurlburt, 1986). By considering the flow to be two-layered, it is possible to modify the RG-SI formalism to include a topography-related term in the regularization.

As all the presented results were established using simulated data, the next step would be to apply the RG-SI method to real observation. Although the eNATL60 simulation is realistic enough to grant confidence into the model's ability to handle real data, moving to such data yields notable challenges. Especially, initial and boundary conditions on the surface stream function would have to be addressed, whether estimated through climatology or gridded products, or included into the control vector.

This new method highlights the benefits of deriving model error parameterization from the general physical framework of the model. The use of physics-based regularization has also proven relevant, especially in the absence of any prior estimate of the parameterized field, which can be very useful when there are no direct observations of the true state of the field. Successfully applying RG-SI to real SWOT data would be a significant step forward, assessing the relevancy of the method in a real-world application.

A RG-SI sublayer parameterization

Gaussian envelopes' standard deviation are chosen so that consecutive envelopes cross at $-3\,\mathrm{dB}$. Hence, considering consecutive time-related (resp., space-related) envelopes to be distant from Δt (resp., Δx and Δy), their standard deviations are given by

$$\sigma_T = \frac{\Delta t}{\sqrt{2\ln 2}}, \tag{26a}$$

$$\sigma_X = \frac{\Delta x}{\sqrt{2\ln 2}}, \tag{26b}$$

$$\sigma_Y = \frac{\Delta y}{\sqrt{2\ln 2}}. \tag{26c}$$

B RG-F implementation

RG-F's forcing is parameterized using a wavelet-like decomposition. Support functions span over different scales, both in time and space, and are built so that large spatial patterns are associated to large temporal patterns. Implementation is inspired from the methodology of Le Guillou et al., 2025. Wavelet functions are defined using cosine and sine functions, windowed using a Gaussian envelope (both in time and space). Since these cosine and sine functions are associated with different wave numbers, envelope widths are adjusted accordingly (standard deviation is computed from the inverse of the corresponding wave number). Lower frequency functions are thus windowed by broader envelopes.

Regarding the regularization, we penalize deviation from the 0 for the forcing. Supposing that we have considered N different scales, since each decomposition coefficient weight a single basis function, it can be matched to its corresponding scale. Regrouping the coefficient by similar scale yields N sets of coefficients, $(\theta_1 \in \mathbb{R}^{M_1}, \dots, \theta_N \in \mathbb{R}^{M_N})$. We have tried to equally penalize these coefficients by defining

$$\mathcal{J}_{\mathrm{reg}}(\theta) = \sum_{i=1}^{N} \|\theta_i\|^2, \tag{27}$$

but such penalizing was actually preventing the model from accurately reconstructing the surface field. Indeed, forcing was either too strongly constrained and thus hardly able to improve the RG0 model, or too loosely constrained and producing small-scale-intensified fields that resulted in very noisy surface stream function reconstructions. To overcome this issue, we penalize the coefficients according to their scale. Hence, denoting $(k_1, \dots, k_N)$ the typical wave numbers of each scale, we penalize the decomposition coefficients using

$$\mathcal{J}_{\mathrm{reg}}(\theta) = \sum_{i=1}^{N} \left(\frac{k_i}{k_1}\right)^{5/3} \|\theta_i\|^2. \tag{28}$$

The choice of such penalization is inspired by Kolmogorov's theory (Kolmogorov & Levin, 1991). Indeed, since small scales (large k_i) are expected to carry less energy, we increase the strength of the regularization for such scales following a 5/3 power law. Using this

regularization has improved the RG-F reconstruction capacities while producing smoother fields. Relative weighting of the regularization loss in comparison to the observation loss is then adjusted using the hyperparameter γ as introduced in eq. (18).

C Projected three-layer QG sublayer stream function performances

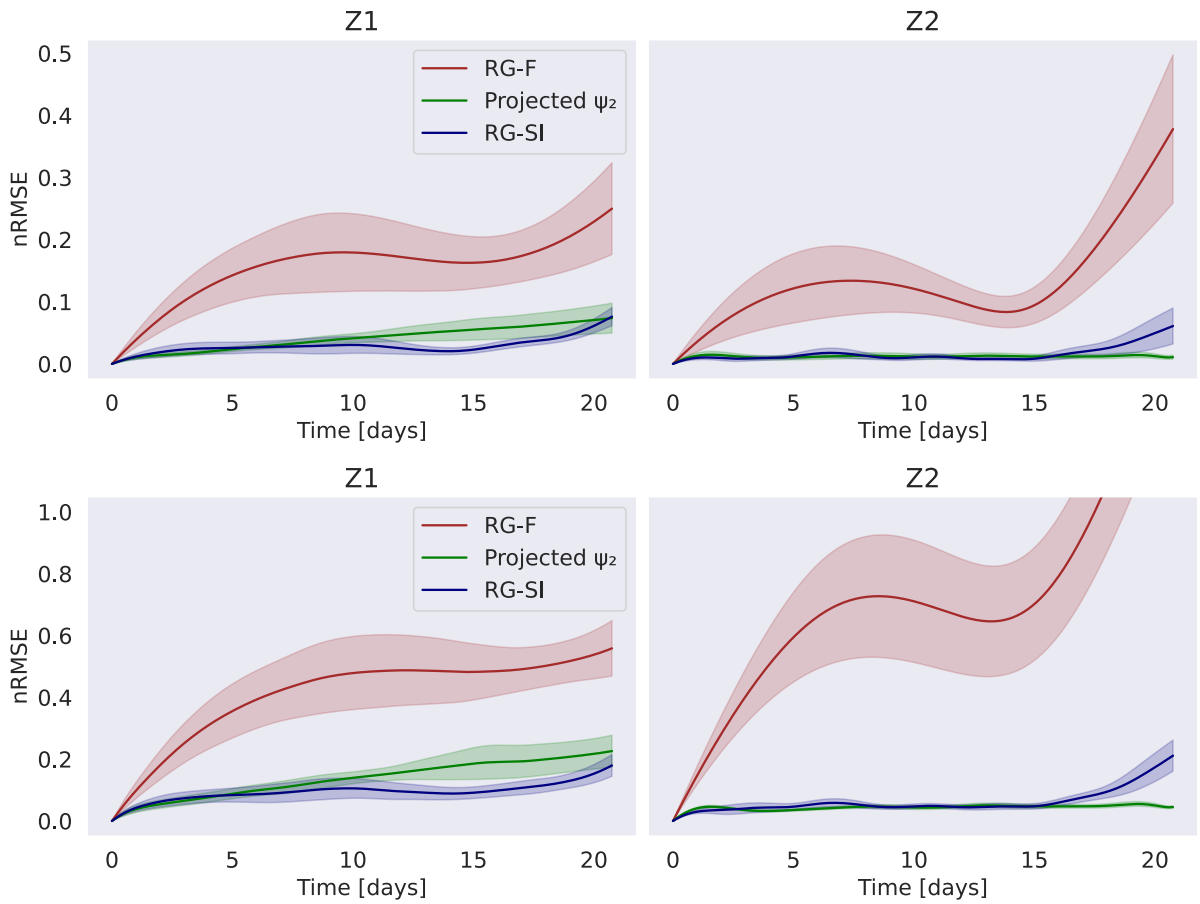

Figure C.1: nRMSE on surface stream function and velocities (averaged over 12 20-days periods, each separated by 20 days) on Z1 (left) and Z2 (right). Standard RG-SI model is identified by the solid blue line, RG-SI with coefficient obtained through the projection of ψ_2 (from the reference three-layer QG model) onto the reduced order space is shown by the green line and RG-F by the brown one. Y-axis is shared by both panels.

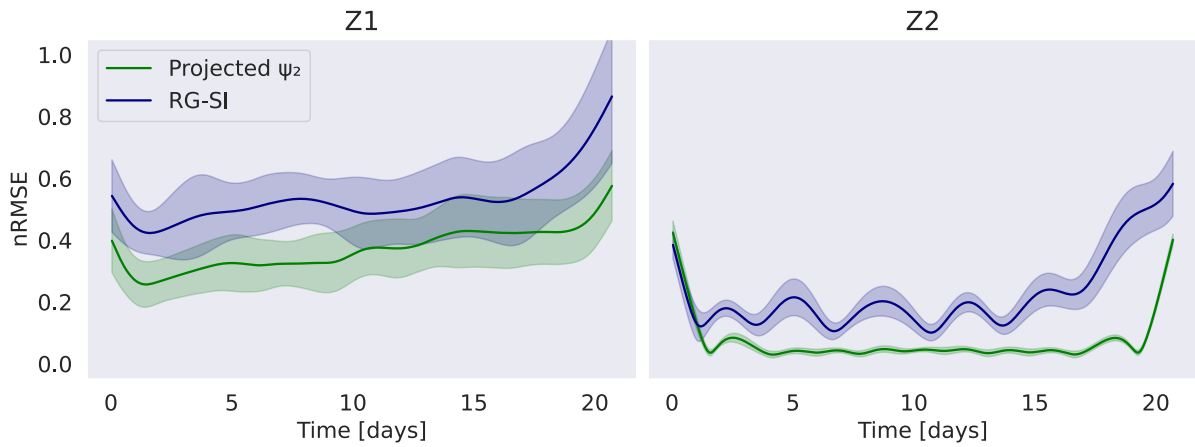

Figure C.2: nRMSE on material derivative of three-layer QG sublayer stream function (averaged over 12 20-days periods, each separated by 20 days) on Z1 (left) and Z2 (right). Standard RG-SI model is identified by the solid blue line whereas RG-SI with coefficient obtained through the projection of ψ_2 (from the reference three-layer QG model) onto the reduced order space is shown by the green line. Y-axis is shared by both panels.

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